Improved guinea-pig ventricular cell model incorporating a diadic space, I_{Kr} and I_{Ks}, and length- and tension-dependent processes

Denis Noble PhD FRS Hon FRCP, Anthony Varghese PhD, Peter Kohl MD PhD, Penelope Noble BSc MSc

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The guinea-pig ventricular cell model, originally developed by Noble et al in 1991, has been greatly extended to include accumulation and depletion of calcium in a diadic space between the sarcolemma and the sarcoplasmic reticulum where, according to contempory understanding, the majority of calcium-induced calcium release is triggered. The calcium in this space is also assumed to play the major role in calcium-induced inactivation of the calcium current. Delayed potassium current equations have been developed to include the rapid (I_{Kr}) and slow (I_{Ks}) components of the delayed rectifier current based on the data of of Heath and Terrar, along with data from Sanguinetti and Jurkiewicz. Length- and tension-dependent changes in mechanical and electrophysiological processes have been incorporated as described recently by Kohl et al. Drug receptor interactions have started to be developed, using the sodium channel as the first target. The new model has been tested against experimental data on action potential clamp, and on force-interval and duration-interval relations; it has been found to reliably reproduce experimental observations.

Key Words: *Computer model, Delayed potassium current, Diadic space, Guinea-pig ventricle*

Amélioration d'un modèle de cellule ventriculaire de cobaye intégrant un espace diadique, I_{Kr} et I_{Ks} et des procédés dépendant de la durée et de la tension

RÉSUMÉ : Le modèle de cellule ventriculaire du cobaye mis au point originalement par Noble et coll. en 1991, a été considérablement étendu afin d'inclure l'accumulation et l'épuisement de calcium dans un espace diadique entre le sarcolemme et le réticulum sarcoplasmique où, selon les connaissances actuelles, est déclenchée la majeure partie de la libération du calcium induite par le calcium. Le calcium dans cet espace jouerait également un rôle de premier plan dans l'inactivation des courants calciques induite par le calcium. Les équations actuelles impliquant le potassium retard ont été mis au point afin d'inclure les composantes rapides (I_{Kr}) et lentes (I_{Ks}) du courant compensateur retard sur la base des données de Heath et Terrar avec les données de Sanguinetti et Jurkiewicz. Les changements dépendant de la durée et de la tension sur le plan des procédés mécaniques électrophysiologiques ont été incorporés comme l'ont récemment décrit Kohl et coll. Les interactions des récepteurs des médicaments ont commencé à être mises au point à l'aide des canaux sodiques comme première cible. Le nouveau modèle a été mis à l'épreuve avec les données expérimentales sur le potentiel d'action et sur les relations entre l'intervalle de force et l'intervalle de durée. Ce modèle permet de reproduire avec fiabilité les observations expérimentales.

University Laboratory of Physiology, University of Oxford, Oxford, United Kingdom
Correspondence and reprints: Prof Denis Noble, University Laboratory of Physiology, University of Oxford, Parks Road, Oxford OX1 3PT, United Kingdom. Fax 44-1865-272554, e-mail denis.noble@physiol.ox.ac.uk

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The guinea-pig ventricular cell model incorporated into the OXSOFT HEART program was originally constructed to analyze the role of sodium-calcium exchange during the cardiac action potential (1). This was also the primary purpose of the atrial cell models (2,3) from which it borrowed many features. The ventricular model has, however, been widely used in many other applications; it is also being used as the generic model for a variety of other species, including rat (4) and the human (unpublished data). In these developments, a number of important processes have been added to the model. The purpose of this paper is to describe these developments and to test the developed model against experimental data.

An important criterion in the design of the new model is that it should reproduce additional physiological functions without greatly increasing computation times. One of the strategic aims of our work is to incorporate cell models into large networks (5,6) and eventually whole heart simulations, in which cell models, as described here, form a basis for a much more ambitious project. Moreover, to reconstruct pathological conditions such as ischemia, many other cellular processes are being incorporated, such as cell metabolism (7) and further drug-receptor interactions. Speed of computation is therefore very important. For this reason, the modelling of the diadic space described here does not compete with more complete attempts to model diffusion and binding of calcium within such a space, such as those of Langer and Peskoff (8) and Jafri et al (9). The results of more complete modelling have, rather, been used to help design simpler, computationally efficient models.

MATERIALS AND METHODS

The model has been developed with the use of OXSOFT HEART 4.8 (Oxsoft Ltd, Oxford, United Kingdom) as the starting point. The full list of equations for the improved model is given in the Appendix. A more qualitative account of the rationale and main new features of the modelling follows.

Diadic space: There is strong evidence to suggest that a large fraction of the calcium entering the cell through L-type calcium channels ($I_{Ca,L}$) first increases calcium concentration in a small space between the sarcolemma and the sarcoplasmic reticulum (SR). This arrangement has functional advantages because it ensures a rapid and large rise of calcium concentration to initiate calcium-induced release of calcium from the SR. This process has been modelled in the simplest possible way. A fraction of the intracellular volume (Vol_{ds}) is assumed to be occupied by the diadic space. In the simulations shown here, the space was set to be 10% of the total cell volume, but the results are essentially similar with spaces of 1% and 0.1%, provided that allowance is made for the higher concentrations of calcium achieved by altering the Michaelis constant (K_m) for binding to the calcium release sites. A suitable value for this constant ($K_{m,Ca,ds}$) is 0.01 when Vol_{ds} is set to 0.1. This formulation immediately removes a major problem encountered with the previous model, which is that the K_m for calcium necessary to achieve calcium release

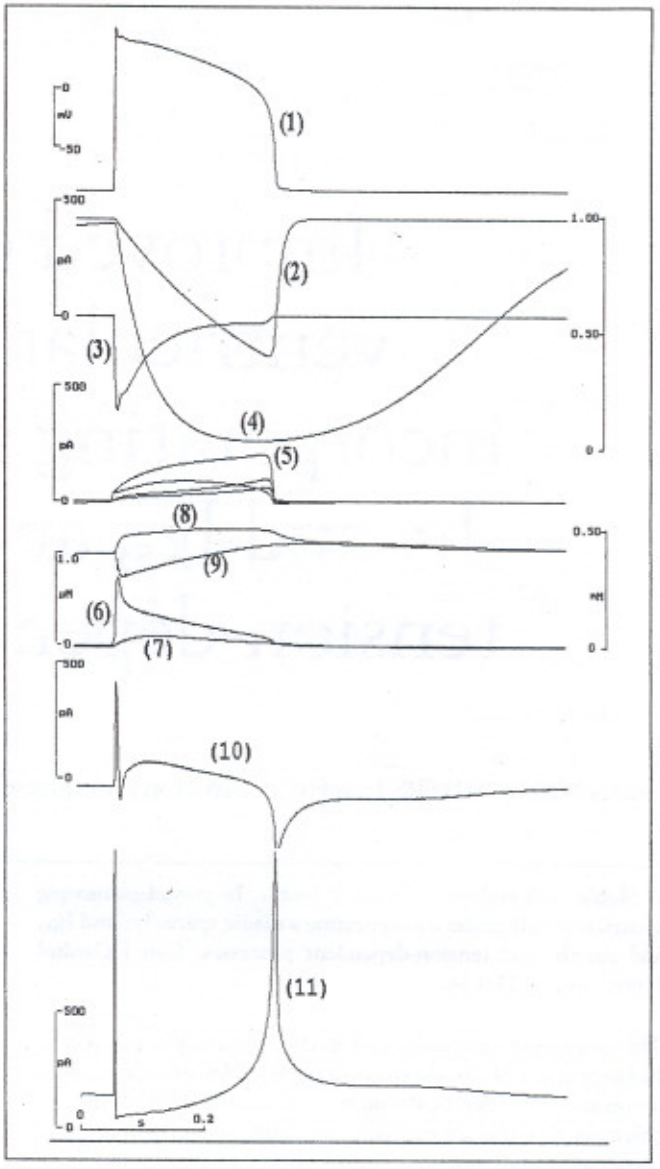


Figure 1) *Components of the guinea-pig ventricular cell action potential model. Trace (1) is membrane potential. Trace (2) is voltage-dependent inactivation of L-type calcium channel current ($I_{Ca,L}$) (right scale). Trace (3) is $I_{Ca,L}$ (left scale). Trace (4) is calcium-dependent inactivation of $I_{Ca,L}$ (right scale). Trace (5) is delayed rectifier current (I_K), incorporating its components: I_K total, top trace; slowly activating component (I_{Ks}), middle trace; fast decay (I_{Kr1}) (above) and slow decay (I_{Kr2}) (below) components of rapidly activating component, bottom trace. Trace (6) is intracellular calcium concentration (μM scale). Trace (7) is diadic space calcium concentration (mM scale). Trace (8) is calcium concentration in sarcoplasmic reticulum (SR) uptake store (mM scale). Trace (9) is calcium concentration in SR release store (mM scale). Trace (10) is sodium-calcium exchanger. Trace (11) is inward rectifier current*

through a rise in bulk cytosol calcium concentration was relatively low (a value around 0.005 mmol was required in the 1991 model) and this, in turn made the system too sensitive to calcium overload. The exact fractions of calcium channel ($Fract I_{Ca,L}$) and sodium-calcium exchanger ($Fract I_{NaCa}$) proteins contributing to the diadic space calcium change are not known with any accuracy. For simplicity here it is as-

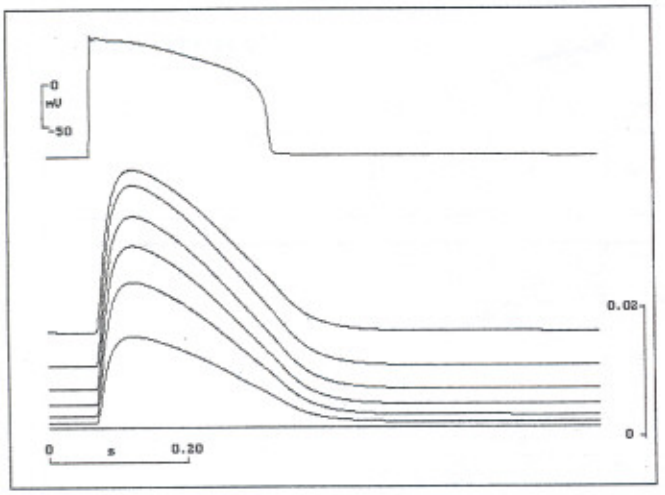


Figure 2) *Effect of sarcomere length (SL) on isometric tension. Upper trace is membrane potential. Lower traces are the isometric tension curves (arbitary units), at values of SL from 1 to 2.2 μm, in steps of 0.2 μm*

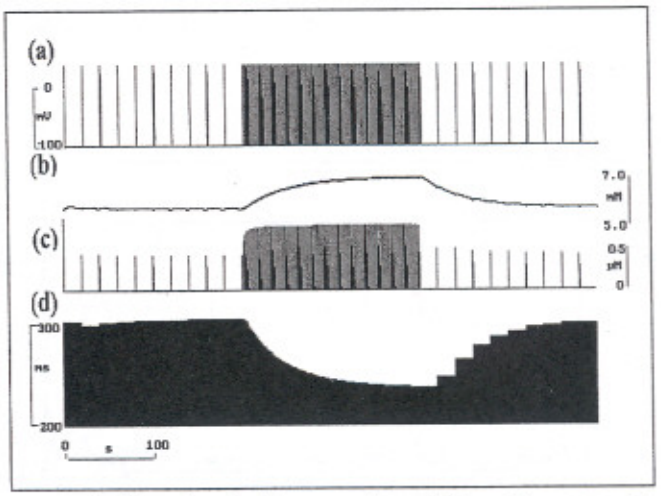


Figure 3) *Effect of stimulus frequency. Trace (a) is membrane potential. Trace (b) is intracellular sodium concentration. Trace (c) is intracellular calcium concentration. Trace (d) is action potential duration. Stimulus frequency from 2 to 20 Hz and back to 2 Hz*

sumed that Fract$I_{Ca,L}$=1 and FractI_{NaCa}=0.001, so that all the calcium channels flow into the space, but only a very small fraction of the exchanger does so. However, these fractions may be modified by the user. Calcium in the diadic space then equilibrates with the free cytosol calcium with a rate constant that was usually set to 10 (ie, a time constant of 100 ms). This corresponds to the relatively slow equilibration that emerges from the Langer and Peskoff model as a consequence of the presence of anionic calcium binding sites. Modelling of the release of calcium from the SR into this space has not, however, been attempted. It is assumed for simplicity that activation of calcium release sites by the diadic space calcium triggers calcium release in the whole cytosol.

Finally, calcium in the diadic space is assumed to bind to inhibitory sites on the calcium channels with a K_m usually set to 0.01 mmol and with a rate constant (R_{decay}) set to 20. As shown in Figure 1, these parameters generate a calcium current time course during the action potential that reproduces the main features of the action potential clamp experimental data of Linz and Meyer (10). A 10-fold decrease in Vol_{ds} (which results in an earlier increase in diacid space calcium concentration, a decrease in $I_{Ca,L}$ inactivation and a decrease in action potential duration) can be compensated for by a 10-fold increase in $K_{m,Ca,ds}$ and R_{decay}. This has negligible effect on intracellular calcium concentration, SR uptake and SR release.

Components of the delayed rectifier current: Modelling of the components of the delayed rectifier current (I_K) was based on the experimental data of Heath and Terrar (11,12) along with data from Sanguinetti and Jurkiewicz (13). These data suggest that there are two components, based on the constants of activation and deactivation as well as pharmacological profiles: a rapidly activating component (I_{Kr}) is sensitive to E-4031 and insensitive to thiopentone and propafol, while a slowly activating component (I_{Ks}) is insensitive to E-4031 and sensitive to both thiopentone and propafol. The deactivation kinetics match those of Gintant (14) and suggest that I_{Kr} has a biexponential decay, represented here by I_{Kr1} for the fast and I_{Kr2} for the slow component. The slower of the two decay rates, I_{Kr2}, is slower than the deactivation rate of I_{Ks}. The equations are listed as equations [7] to [9] and [30] to [32] in the Appendix, and the behaviour of the components during the computed action potential is shown in Figure 1.

Contraction, tension, and stretch-dependent and tension-dependent processes: The Hilgemann-Noble (2) and Noble et al (1) models already incorporate a description of calcium binding to troponin and the subsequent events leading to cross-bridge formation. This work has been built upon by the addition of several features:

- ***Computation of isometric tension:*** Equations have been incorporated to calculate isometric tension as a function of sarcomere length. The equations are purely descriptive, though as Figure 2 shows they succeed in giving a good description of the changes in resting and active tension as cell length is increased from 1 μm, below which tension is assumed to be zero, to 2.2 μm. Equations [23] to [28] in the Appendix describe this modelling.
- ***Cellular mechanisms of cardiac mechano-electric feedback:*** These are fully described by Kohl et al (15) in this issue. In short, selective and nonselective stretch-activated ion conductances have been introduced, whose activation may be linked to changes either in sarcomere length or in isometric tension (equations are listed in the Appendix as [54] to [59]).This, together with length/tension-dependent modulation of calcium binding to troponin (equations [19] and [19a]) and a mechanically modulated sarcoplasmic leak current (equations [52] and [52a]),

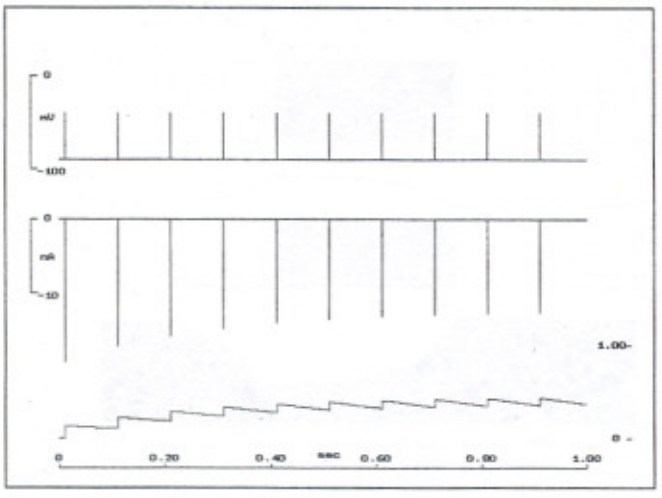


Figure 4) *Time-dependent block of sodium current (I_{Na}) by mexiletine (100 μmol). Upper trace is voltage: 0.6 ms pulse to –40 mV from holding potential of –90 mV repeated every 100 ms. Middle trace is I_{Na}. Bottom trace is the fraction of sodium channels blocked by the drug*

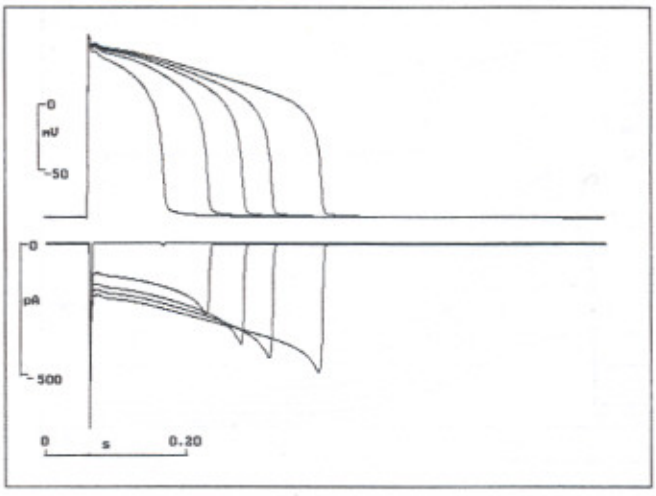


Figure 5) *Effect of the persistent sodium current on action potential. Upper traces are membrane potential. Lower traces are the total sodium current. Values of persistent sodium current conductance are 0, 0.0025, 0.0035, 0.004 and 0.0045 μS*

allows the reproduction of numerous experimental findings and may help to explain some of the diversity in the responses to mechanical stimulation observed under different experimental conditions.

Drug-receptor interactions: Although block of channels and transporters by drugs can be studied by varying parameters such as conductance, it is well known that a number of drugs have kinetics with the same time scale as the cardiac action potential. The modulated receptor model of Hondeghem and Katzung (16) of sodium channel block was incorporated as an example of a nontrivial block of an ionic mechanism by drugs (equations [3a], [65] and [66]).

RESULTS

Force interval, duration interval relations: The equations for uptake and release of calcium to and from the SR were originally developed by Hilgemann and Noble to reconstruct the behaviour of rabbit atrial tissue. The force interval and duration intervals in atrial and ventricular tissue differ very substantially (17). Some of the key parameters have therefore been adjusted to give results similar to those obtained in ventricular cells. Figure 3 shows that, with these adjustments, the model is capable of reproducing the positive contraction staircase that occurs during increases in frequency of stimulation in guinea-pig. As in the experimental recordings of Harrison and Boyett (18), this is accompanied by changes in intracellular sodium, which, via effects on sodium-calcium exchange, also produce changes in action potential shape and duration.

Drug-receptor interactions: When a train of voltage clamp pulses is applied at a constant rate in the presence of a sodium channel blocker, the sodium current is seen to decay exponentially to a pseudosteady state value. Figure 4 shows use-dependent block by 100 μmol mexiletine under voltage clamp protocol.

The persistent sodium current: Although not extensively characterized in guinea-pig, the persistent sodium current has been shown to be present in rat ventricular myocytes (19). Figure 5 shows the hypothetical effect on action potential duration of progressive increase in the conductance of this current ($G_{p,Na}$). The potential effect even of very small (compared with the fast sodium conductance responsible for the action potential upstroke) amounts of $G_{p,Na}$ is clearly very large. This highlights the importance of further experiments to determine the magnitude of $G_{p,Na}$ in the guinea-pig and in other species with long action potentials. The effects are considerably larger than expected in animals, such as the rat, with short ventricular action potential plateaus.

Action potential clamp: The action potential clamp technique involves using a prerecorded action potential as the voltage clamp waveform so that the membrane potential is forced to follow the action potential even when components of ionic current that would normally contribute to generating this time course are modified. By selective blocking of different current components it is therefore possible to determine the time course of each current during the action potential. This provides a powerful technique for validation of models that attempt to predict these time courses from equations based on kinetics measured during step clamp protocols. Such validation is necessary for two reasons. First, there is no guarantee that the data obtained from step clamp protocols will necessarily succeed in predicting current time courses during other waveforms because the history of membrane potential change may also be important. When processes such as calcium-dependent inactivation of the calcium current are included, such history dependence is in fact very likely. Second, the interactions among all the processes involved in generating the overall electrical behaviour are complex. Even small errors in one component may in turn perturb the behaviour of others. Experimental data from action potential clamping therefore provide a welcome form of additional

validation. The ionic current time courses generated by the present model do in fact reproduce the main features of action potential clamp experiments (20-23).

For the purpose of future work with the OXSOFT HEART program, the action potential clamp has now also been incorporated as a facility in the program. Previously computed action potentials, or indeed any other computed waveform, can be used to control the voltage clamp mode within the program.

CONCLUSIONS

This paper reports a major set of updates of the 1991 ventricular cell model. The range of applicability of the model has, in consequence, been greatly extended, as indicated also by other papers in this issue (15,24). As a whole, these studies demonstrate the accuracy of the reconstructions now possible.

One of the main motivations in developing the new model was that it should be capable of reconstructing the experimental data emerging from use of the action potential clamp technique. This is a powerful tool not only in experimental investigations but also in modelling because it provides validation from the 'top' end of a system that is built on the basis of the description of ionic currents. This is not trivial. It is no longer sufficient simply to reproduce action potential shapes well (different 'bottom-up' models achieve that); it is crucial to reproduce the actual time courses of current changes when an action potential is used as the voltage clamp signal. The present model is capable of doing this.

Finally, an important requirement was that the increased range of applicability should be secured with minimal increase in computing time. This is the case. Using a 133 MHz pentium machine, the 1991 model required 10 s to compute 800 ms of activity. The new model, despite the many additional computed features, required only 11 s for the same period of activity. The increase in computation time has therefore been limited to about 10%. This performance makes the model well suited for incorporation into large network and whole heart computations and, given the rapid increase in computing power, provides plenty of scope for the incorporation of biochemical processes.

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APPENDIX
SINGLE GUINEA-PIG VENTRICULAR CELL MODEL EQUATIONS

VELOCITY FIELD EQUATIONS

$$\dot{V} = \frac{-1}{C_m}\left(\begin{array}{l} I_{K1} + I_K + I_{K,ATP} + I_{Ca,L,K} + I_{b,K} + I_{NaK} + I_{Na} + I_{b,Na} + I_{p,Na} + \\ I_{Ca,L,Na} + I_{NaCa} + I_{NaCa,ds} + I_{Ca,L,Ca} + I_{Ca,L,Ca,ds} + I_{Ca,L,K,ds} + I_{Ca,L,Na,ds} + \\ I_{b,Ca} + I_{Na\text{ - stretch}} + I_{K\text{ - stretch}} + I_{Ca\text{ - stretch}} + I_{Ns\text{ - stretch}} + I_{An\text{ - stretch}} + I_{K,ACh} \end{array}\right) \quad [1]$$

$$\dot{m} = \frac{200(V+41)}{1-e^{-0.1(V+41)}}(1-m) - 8000e^{-0.056(V+66)}m \quad [2]$$

$$\dot{h} = 20e^{-0.125(V+75)}(1-h) - \frac{2000}{1+320e^{-0.1(V+75)}}h \quad [3]$$

For 3(a) see end of listing

$$\dot{d} = \frac{90(V+19)}{1-e^{-(V+19)/4}}(1-d) - \frac{36(V+19)}{e^{(V+19)/10}-1}d \quad [4]$$

$$\dot{f} = \frac{12}{1+e^{-(V+34)/4}}\left(\frac{119[Ca^{2+}]_i}{K_{cacoff}+[Ca^{2+}]_i}+1\right)(1-f) - \frac{6.25(V+34)}{e^{(V+34)/4}-1}f \quad [5]$$

$$\dot{f}_{ds} = \frac{12}{1+e^{-(V+34)/4}}\left(\frac{119[Ca^{2+}]_{ds}}{K_{dsoff}+[Ca^{2+}]_{ds}}+1\right)(1-f_{ds}) - \frac{6.25(V+34)}{e^{(V+34)/4}-1}f_{ds} \quad [6]$$

$$\dot{x}_{r1} = \frac{50}{1+e^{-(V-5)/9}}(1-x_{r1}) - 0.05e^{\frac{(V-20)}{15}} \quad [7]$$

$$\dot{x}_{r2} = \frac{50}{1+e^{-(V-5)/9}}\left(1-x_{r2}\right) - 0.4e^{-\left(\frac{V+30}{30}\right)^3} \quad [8]$$

$$\dot{x}_s = \frac{14}{1+e^{-\frac{(V-40)}{9}}}\left(1-x_s\right) - e^{\frac{-V}{45}}x_s \quad [9]$$

$$\dot{x}_{ACh} = \alpha_{ACh}\left(1-x_{ACh}\right) - \beta_{ACh}.x_{ACh} \quad [10]$$

$$\left[\dot{Na}^+\right]_i = \frac{-1}{V_i F}\left(I_{Na} + I_{p,\,Na} + I_{b,\,Na} + 3I_{NaK} + 3I_{NaCa} + 3I_{NaCa,\,ds} + I_{Ca,\,L,\,Na} + I_{Na\,-\,stretch}\right) \quad [11]$$

$$\left[\dot{K}^+\right]_i = \frac{-1}{V_i F}\left(I_{K1} + I_{Ca,\,L,\,K} + I_{b,\,K} + I_K - 2I_{NaK} + I_{K\,-\,stretch}\right) \quad [12]$$

$$\left[\dot{Ca}^{2+}\right]_o = \frac{1}{2V_{cell}V_{ecs}F}\left(I_{Ca,\,L,\,Ca} + I_{Ca,\,L,\,Ca,\,ds} + I_{b,\,Ca} - 2I_{NaCa} + I_{Ca\,-\,stretch}\right) - Diff_{Ca}\left(\left[Ca^{2+}\right]_o - \left[Ca^{2+}\right]_b\right) \quad [13]$$

$$\left[\dot{Ca}^{2+}\right]_i = \frac{-1}{2V_i F}\left(I_{Ca,\,L,\,Ca} + I_{b,\,Ca} - 2I_{NaCa} + I_{Ca\,-\,stretch}\right) - I_{up} + I_{rel}\frac{V_{SRup}V_{rel}}{V_i V_{up}} - \left[Ca^{2+}\right]_{cal\,mod} - \left[Ca^{2+}\right]_{troponin} + \frac{\left[Ca^{2+}\right]_{ds}}{\left[Ca^{2+}\right]_{ds} + K_{decay}}R_{decay} \quad [14]$$

$$\left[\dot{Ca}^{2+}\right]_{ds} = \frac{-1}{2V_{ds}F}\left(I_{Ca,\,L,\,Ca,\,ds} - 2I_{NaCa,\,ds}\right) - \frac{\left[Ca^{2+}\right]_{ds}}{\left[Ca^{2+}\right]_{ds} + K_{decay}}R_{decay} \quad [15]$$

$$\left[\dot{Ca}^{2+}\right]_{up} = \frac{V_i}{V_{SRup}}I_{up} - I_{tr} \quad [16]$$

$$\left[\dot{Ca}^{2+}\right]_{rel} = \frac{V_{up}}{V_{rel}}I_{tr} - I_{rel} \quad [17]$$

$$\left[\dot{Ca}^{2+}\right]_{cal\,mod} = 10^5\left(M_{trop} - \left[Ca^{2+}\right]_{cal\,mod}\right)\left[Ca^{2+}\right]_i - 50\left[Ca^{2+}\right]_{cal\,mod} \quad [18]$$

$$\left[\dot{Ca}^{2+}\right]_{troponin} = \alpha_{trop}e^{\gamma_{trop\,SL}SL}\left(C_{trop} - \left[Ca^{2+}\right]_{troponin}\right)\left[Ca^{2+}\right]_i - \beta_{trop}\left[Ca^{2+}\right]_{troponin} \quad [19]$$

OR

$$\left[\dot{Ca}^{2+}\right]_{troponin} = \alpha_{trop}e^{\gamma_{trop\,IT}IT}\left(C_{trop} - \left[Ca^{2+}\right]_{troponin}\right)\left[Ca^{2+}\right]_i - \beta_{trop}\left[Ca^{2+}\right]_{troponin} \quad [19a]$$

$$\dot{f}_{activator} = \left(1 - f_{activator} - f_{product}\right)\left(500\left(x_{SRRel}\right)^2 + 600e^{(V-40)0.08}\right) - f_{activator}\left(500\left(x_{SRRel}\right)^2 + 60\right) \quad [20]$$

$$\dot{f}_{product} = f_{activator}\left(500\left(x_{SRRel}\right)^2 + 60\right) - f_{product} \quad [21]$$

$$\dot{f}_{light-chain} = k_1 X_1^{\,2} X_2\left(1 - f_{light-chain}\right) - k_2 f_{light-chain} \quad [22]$$

$$\dot{f}_{crossbridge} = k_3 f_{light-chain}\left(1 - f_{crossbridge}\right) - k_4 f_{crossbridge} \quad [23]$$

$$T_{rest} = 0.0002e^{2SL} \quad [24]$$

where T_{rest} = resting tension and SL = sarcomere length in μm

$$T_{act} = 1 - e^{-3(SL-1)} \text{ for } SL > 1 \quad [25a]$$

$$T_{act} = 0 \text{ for } SL \leq 1 \quad [25b]$$

$$\text{Overlap} = 1 \text{ for } SL < 2 \quad [26a]$$

$$\text{Overlap} = 1 - 0.625(SL - 2) \text{ for } SL > 2 \quad [26b]$$

$$CB_{avail} = T_{act}.\text{Overlap}.[CB] \quad [27]$$

$$T_{total} = f_{crossbridge}.CB_{avail} + T_{rest} \quad [28]$$

MEMBRANE CURRENT EQUATIONS

$$I_{K1} = G_{K1}\frac{[K^+]_o}{[K^+]_o + k_{mk1}}\left(\frac{V - E_K}{1 + e^{\frac{V - E_K - 10}{RT/2F}}}\right) \quad [29]$$

$$I_K = I_{Kr} + I_{Ks} + I_{K,Na} \quad [30]$$

$$I_{Kr} = \left(G_{Kr1}X_{r1} + G_{Kr2}X_{r2}\right)\frac{1}{1 + e^{\frac{V+9}{22.4}}}(V - E_K) \quad [31]$$

$$I_{Ks} = G_{Ks}X_s^{\,2}(V - E_{Ks}) \quad [32]$$

$$I_{b,K} = G_{bk}(V - E_K) \quad [33]$$

$$I_{K,ATP} = \frac{G_{K.ATP}(V + 80)}{1 + \left(\frac{[ATP]}{K_{ATP}}\right)^2} \quad [34]$$

$$I_{K,Na} = G_{K,Na}\frac{[Na^+]_i}{[Na^+]_i + k_{kna}}(V - E_K) \quad [35]$$

$$I_{Na} = G_{na}(V - E_{mh})m^3h \quad [36]$$

$$I_{p,Na} = G_{pna}\frac{1}{1+e^{\left(\frac{V+52}{8}\right)}}(V - E_{Na}) \quad [37]$$

$$I_{b,Na} = G_{bna}(V - E_{Na})\frac{[Na^+]_i}{140} \quad [38]$$

$$I_{Ca,L,Ca} = 4P_{Ca}d(1-f)\frac{k_{cachoff}}{k_{cachoff}+[Ca^{2+}]_i}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/2F}}}\left[[Ca^{2+}]_i e^{\frac{50}{RT/2F}} - [Ca^{2+}]_o e^{\frac{-(V-50)}{RT/2F}}\right] \quad [39]$$

$$I_{Ca,L,K} = P_{CaK}P_{Ca}d(1-f)\frac{k_{cachoff}}{k_{cachoff}+[Ca^{2+}]_i}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[[K^+]_i e^{\frac{50}{RT/2F}} - [K^+]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad [40]$$

$$I_{Ca,L,Na} = P_{CaNa}P_{Ca}d(1-f)\frac{k_{cachoff}}{k_{cachoff}+[Ca^{2+}]_i}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[[Na^+]_i e^{\frac{50}{RT/2F}} - [Na^+]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad [41]$$

$$I_{Ca,L,Ca,ds} = 4P_{Ca}d(1-f_{ds})\frac{k_{dsoff}}{k_{dsoff}+[Ca^{2+}]_{ds}}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/2F}}}\left[[Ca^{2+}]_{ds} e^{\frac{50}{RT/2F}} - [Ca^{2+}]_o e^{\frac{-(V-50)}{RT/2F}}\right] \quad [42]$$

$$I_{Ca,L,K,ds} = P_{CaK}P_{Ca}d(1-f_{ds})\frac{k_{dsoff}}{k_{dsoff}+[Ca^{2+}]_{ds}}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[[K^+]_i e^{\frac{50}{RT/2F}} - [K^+]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad [43]$$

$$I_{Ca,L,Na,ds} = P_{CaNa}P_{Ca}d(1-f_{ds})\frac{k_{dsoff}}{k_{dsoff}+[Ca^{2+}]_{ds}}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[[Na^+]_i e^{\frac{50}{RT/2F}} - [Na^+]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad [44]$$

$$I_{b,Ca} = G_{bca}(V - E_{Ca}) \quad [45]$$

$$I_{K,ACh} = G_{KACh}\frac{[K^+]_o}{[K^+]_o + k_{mk1}}x_{ACh}\frac{1}{1+\left(\frac{k_D}{[ACh]}\right)^2}\frac{V - E_K}{1+e^{\frac{(V-E_K-10)}{RT/2F}}} \quad [46]$$

PUMP/EXCHANGER CURRENT EQUATIONS

$$I_{NaK} = I_{NaKmax} \frac{[K^+]_o}{[K^+]_o + k_{mk}} \frac{[Na^+]_i}{[Na^+]_i + k_{mNa}} \quad [47]$$

$$I_{NaCa} = k_{NaCa} \frac{e^{\gamma \frac{V}{RT/F}} [Na^+]_i^3 [Ca^{2+}]_o - e^{-(1-\gamma)\frac{V}{RT/F}} [Na^+]_o^3 [Ca^{2+}]_i}{1 + d_{NaCa}\left([Ca^{2+}]_i [Na^+]_o^3 + [Ca^{2+}]_o [Na^+]_i^3\right)} \quad [48]$$

$$I_{NaCa,\,ds} = k_{NaCa} \frac{e^{\gamma \frac{V}{RT/F}} [Na^+]_i^3 [Ca^{2+}]_o - e^{-(1-\gamma)\frac{V}{RT/F}} [Na^+]_o^3 [Ca^{2+}]_{ds}}{1 + d_{NaCa}\left([Ca^{2+}]_{ds} [Na^+]_o^3 + [Ca^{2+}]_o [Na^+]_i^3\right)} \quad [49]$$

CALCIUM SEQUESTRATION EQUATIONS

$$I_{up} = \frac{3[Ca^{2+}]_i - 0.23[Ca^{2+}]_{up} \frac{k_{cyca} k_{xcs}}{k_{srca}}}{[Ca^{2+}]_i + [Ca^{2+}]_{up} \frac{k_{cyca} k_{xcs}}{k_{srca}} + k_{cyca} k_{xcs} + k_{cyca}} \quad [50]$$

$$I_{tr} = 50\left([Ca^{2+}]_{up} - [Ca^{2+}]_{rel}\right) \quad [51]$$

$$I_{rel} = \left(\frac{f_{activator}}{f_{activator} + 0.25}\right)^2 k_{mca2} [Ca^{2+}]_{rel} + J_{SR-leak} e^{\gamma_{SR,\,SL} SL} \quad [52]$$

OR

$$I_{rel} = \left(\frac{f_{activator}}{f_{activator} + 0.25}\right)^2 k_{mca2} [Ca^{2+}]_{rel} + J_{SR-leak} e^{\gamma_{SR,\,IT} IT} \quad [52a]$$

$$x_{SRRel} = \left(\frac{[Ca^{2+}]_i}{[Ca^{2+}]_i + k_{mca}}\right) + \left(1 - \frac{[Ca^{2+}]_i}{[Ca^{2+}]_i + k_{mca}}\right)\left(\frac{[Ca^{2+}]_{ds}}{[Ca^{2+}]_{ds} + k_{mca,ds}}\right) \quad [53]$$

STRETCH DEPENDENCE

$$f_{stretch} = \frac{1}{1 + e^{-2\gamma_{SAC,SL}\,(SL - SL_{HST})}} \quad [54]$$

OR

$$f_{stretch} = \frac{1}{1 + e^{-2\gamma_{SAC,IT}\,(IT - IT_{HST})}} \quad [54a]$$

$$I_{Ca\text{ - stretch}} = G_{Ca\text{ - stretch}}\, f_{stretch}(V - E_{Ca}) \quad [55]$$

$$I_{K\text{ - stretch}} = G_{K\text{ - stretch}}\, f_{stretch}(V - E_K) \quad [56]$$

$$I_{Na\text{ - stretch}} = G_{Na\text{ - stretch}}\, f_{stretch}(V - E_{Na}) \quad [57]$$

$$I_{Ns\text{ - stretch}} = G_{Ns\text{ - stretch}}\, f_{stretch}(V - E_{Ns\text{ - stretch}}) \quad [58]$$

$$I_{An\text{ - stretch}} = G_{An\text{ - stretch}}\, f_{stretch}(V - E_{An\text{ - stretch}}) \quad [59]$$

REVERSAL POTENTIALS

$$E_{Na} = \frac{RT}{F} \ln\left(\frac{[Na^+]_o}{[Na^+]_i}\right) \quad [60]$$

$$E_K = \frac{RT}{F} \ln\left(\frac{[K^+]_o}{[K^+]_i}\right) \quad [61]$$

$$E_{Ks} = \frac{RT}{F} \ln\left(\frac{[K^+]_o + P_{kna}[Na^+]_o}{[K^+]_i + P_{kna}[Na^+]_i}\right) \quad [62]$$

$$E_{Ca} = \frac{RT}{F} \ln\left(\frac{[Ca^{2+}]_o}{[Ca^{2+}]_i}\right) \quad [63]$$

$$E_{mh} = \frac{RT}{F} \ln\left(\frac{[Na^+]_o + 0.12[K^+]_o}{[Na^+]_i + 0.12[K^+]_i}\right) \quad [64]$$

DRUG/RECEPTOR INTERACTIONS

In order to model the effects of sodium channel blockers with use-dependent kinetics, one existing differential equation has to be changed and two new ones have to be added. Equation [3] becomes:

$$\dot{h} = 20e^{-0.125(V+75)}(1-b-h) - \frac{2000}{1+320e^{-0.1(V+75)}}h + \left(l_r h_D(1-m^3) + l_a m^3 h_D\right) \quad [3a]$$

where h_D is the inactivation gating variable for the drug-bound channels and b is the fraction of channels blocked by the drug. The additional equations are:

$$\dot{h}_D = 20e^{-0.125(V+75+\Delta V)}(b-h_D) - \frac{2000}{1+320e^{-0.1(V+75+\Delta V)}}h_D + \left(k_r h(1-m^3) + k_a m^3 h[D]\right) \quad [65]$$
$$-\left(l_r h_D(1-m^3) + l_a m^3 h_D\right)$$

$$\dot{b} = [D]\left(k_r h(1-m^3) + k_a m^3 h + k_i(1-b-h)\right) - \left(l_r h_D(1-m^3) + l_a m^3 h_D + l_i(1-b-h_D)\right) \quad [66]$$

where k_r, k_i, k_a, l_r, l_i and l_a describe binding kinetics of a drug to the sodium channel and ΔV represents the shift in the inactivation curve induced by the drug. [D] is the concentration of the drug.

Parameters

C_M	95(10)$^{-6}$ μF	$DIFF_{CA}$	5(10)$^{-4}$ NA
		d_{naca}	10^{-4}
$k_{cachoff}$	0.001 mM	γ	0.5
k_{mk1}	10 mM	n_{NaCa}	3
k_{mk}	1 mM		
k_{mNa}	40 mM	k_{cyca}	3(10)$^{-4}$ mM
k_{naca}	10^{-4} nA	k_{xcs}	0.4 mM
k_{mca}	0.001 mM	k_{srca}	0.5 mM
k_1	12,000 mM	F	96485 C/mol
k_2	100 mM	R	8314.41 mJ/(mol °K)
k_3	60 mM	T	310°K
k_4	25 mM		
k_{ATP}	0.1 mM	V_{ecs}	0.4
K_D	0.13 μM ACh	radius	10 μm
k_{kna}	20 mM	length	80 μm
k_{dsoff}	0.001 mM	V_{cell}	10^{-9} π radius^{2}length μL
$k_{mca,ds}$	0.01 mM	V_i	$(1-V_{ecs}-V_{up}-V_{rel})V_{cell}$ μL
k_{decay}	100 ms	V_{ds}	$0.1V_i$ μL
		V_{up}	0.01
R_{decay}	20 mM/s	V_{rel}	0.1
		V_{SRup}	$V_{cell}V_{up}$ μL
I_{NaKmax}	0.14 nA	$J_{SR-leak}$	
		k_{mca2}	250 nA/mM
G_{na}	0.5 μS	M_{trop}	0.02 mM
G_{bk}	0.0017 μS	C_{trop}	0.15 mM
G_{k1}	0.017 μS	α_{trop}	5000/s
G_{bna}	0.00012 μS	β_{trop}	200/s
G_{pna}	0.004 μS	$\gamma_{trop,SL}$ ($\gamma_{trop,IT}$)	2.5
G_{bca}	0.00005 μS	$\gamma_{SR,SL}$ ($\gamma_{SR,IT}$)	1.5
G_{Kr1}	0.0028 μS	$\gamma_{SAC,SL}$ ($\gamma_{SAC,IT}$)	2.5
G_{Kr2}	0.0017 μS		
G_{Ks}	0.0032 μS	$G_{Ca\text{-}stretch}$	0 μS (1 μS)
G_{KACh}	0.00005 μS	$G_{Na\text{-}stretch}$	0 μS (1 μS)
		$G_{K\text{-}stretch}$	0 μS (1 μS)
P_{ca}	0.05 nA/mM	$G_{Ns\text{-}stretch}$	0 μS (1 μS)
P_{cak}	0.002 nA/mM	$G_{An\text{-}stretch}$	0 μS (1 μS)
P_{cana}	0.002 nA/mM	$E_{Ns\text{-}stretch}$	0 mV (–30 mV)
P_{kna}	0.03 nA/mM	$E_{An\text{-}stretch}$	0 mV (–20 mV)
		α_s	2
$[Ca^{2+}]_b$	2 mM	α_v	2
$[K^+]_o$	4 mM	α_{Ach}	0.5/s
$[Na^+]_o$	140 mM	β_{ACh}	0.5/s
$FractI_{Ca,L}$	1	SL_{HST}	2.0 μm
$FractI_{NaCa}$	0.001	IT_{HST}	1.0 au

Units: au Arbitrary units; A Amperes; C Coulomb; J Joules; K Kelvin; M mol/L; S Siemens. Values are expressed as default (where default value is 0, reasonable values are given in brackets)

REFERENCES

1. Noble D, Noble SJ, Bett GCL, Earm YE, Ho WK, So IK. The role of sodium-calcium exchange during the cardiac action potential. Ann NY Acad Sci 1991;639:334-53.
2. Hilgemann DW, Noble D. Excitation-contraction coupling and extracellular calcium transients in rabbit atrium: reconstruction of basic cellular mechanisms. Proc R Soc Lond B Biol Sci 1987;230:163-205.
3. Earm YE, Noble D. A model of the single atrial cell: relation between calcium current and calcium release. Proc R Soc Lond B Biol Sci 1990;240:83-96.
4. Espinosa L, Chouabe C, Noble PJ, Noble D, Bonvallet R. Na^+/Ca^{2+} exchanger is involved in action potentials in hypertrophied rat ventricular cells. J Mol Cell Cardiol (Abst, in press)
5. Winslow RL, Varghese A, Noble D, Adlakha C, Hoytha A. Generation and propagation of ectopic beats induced by spatially localized Na-K pump inhibition in atrial network models. Proc R Soc Lond B Biol Sci 1993;254:55-61.
6. Winslow RL, Cai D, Varghese A, Lai Y-C. Generation and propagation of normal and abnormal pacemaker activity in network models of cardiac sinus node and atrium. Chaos Solitons Fractals 1995;5:491-512.
7. Ch'en FF-T, Vaughan-Jones RD, Clarke K, Noble D. Modelling of internal pH, ion concentration, and bioenergetic changes. Adv Exp Med Biol 1997;430:281-90.
8. Langer GA, Peskoff A. Calcium concentration and movement in the diadic cleft space of the cardiac ventricular cell. Biophys J 1996;70:1169-82.
9. Jafri MS, Rice JJ, Yue DT, Winslow RL. Calcium dynamics in cardiac myocytes. Biophys J 1997;72:A329.
10. Linz KW, Meyer R. Modulation of L-type calcium current by internal potassium in guinea pig ventricular myocytes. Cardiovasc Res 1997;33:110-22.
11. Heath BM, Terrar DA. Separation of the components of the delayed rectifier potassium current using selective blockers of I_{kr} and I_{ks} in guinea-pig isolated ventricular myocytes. Exp Physiol 1996;81:587-603.
12. Heath BM, Terrar DA. The deactivation kinetics of the delayed rectifier components I_{kr} and I_{ks} in guinea-pig isolated ventricular myocytes. Exp Physiol 1996;81:605-21.
13. Sanguinetti MC, Jurkiewicz NK. Two components of cardiac delayed rectifier K^+ current. Differential sensitivity to block by class III antiarrhythmic agents. J Gen Physiol 1990;96:195-215.
14. Gintant GA. Two components of the delayed rectifier current in canine atrium and ventricle. Circ Res 1996;78:26-37.
15. Kohl P, Day K, Noble D. Cellular mechanisms of cardiac mechano-electric feedback in a mathematical model. Can J Cardiol 1998;14:111-9.
16. Hondeghem LM, Katzung BG. Time and voltage dependent interaction of antiarrhythmic drugs with cardiac sodium channels. Biochim Biophys Acta 1977;472:373.
17. Koch-Weser J, Blinks JR. The influence of the interval between beats on myocardial contractility. Pharmacol Rev 1963;15:601-52.
18. Harrison SM, Boyett MR. The role of the Na(+)-Ca(2+) exchanger in the rate-dependent increase in contraction in guinea-pig ventricular myocytes. J Physiol 1995;482:555-66.
19. Saint DA, Ju Y, Gage PW. A persistent sodium current in rat ventricular myocytes. J Physiol 1992;453:219-31.
20. Linz KW, Meyer R. Modulation of L-type calcium current by internal potassium in guinea pig ventricular myocytes. Cardiovasc Res 1997;33:110-22.
21. Doerr Th, Denger R, Trautwein W. Calcium currents in single SA nodal cells of the rabbit heart studied with action potential clamp. Pflügers Arch 1989;413:599-603.
22. Doerr Th, Denger R, Trautwein W. Ionic currents contributing to the action potential in single ventricular myocytes of the guinea pig studied with action potential clamp. Pflügers Arch 1990;416:230-7.
23. Grantham CJ, Cannell MB. Ca^{2+} influx during the cardiac action potential in guinea pig ventricular myocytes. Circ Res 1996;79;194-200.
24. Noble D, Varghese A. Modelling of sodium-overload arrhythmias and their suppression. Can J Cardiol 1998;14:97-100.

Improved Guinea-Pig Ventricular Cell Model Incorporating a Diadic Space, I_{Kr} and I_{Ks}, and Length- and Tension-Dependent Processes

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ERRATA

The L-Type Calcium Current

Equation 6 should be read:-

$$\dot{f}_2 = \left(1 - \frac{\left[Ca^{2+}\right]_i}{K_{cachoff} + \left[Ca^{2+}\right]_i} - f_2\right) \tag{6}$$

or for the diadic space currents:-

$$\dot{f}_{2ds} = R_{decay}\left(1 - \frac{\left[Ca^{2+}\right]_{ds}}{K_{dsoff} + \left[Ca^{2+}\right]_{ds}} - f_{2ds}\right)$$

Equations 39, 40, 41, 42, 43 and 44 should read:-

$$I_{Ca,\,L,\,Ca} = 4P_{Ca}dff_2 \frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/2F}}}\left[\left[Ca^{2+}\right]_i e^{\frac{50}{RT/2F}} - \left[Ca^{2+}\right]_o e^{\frac{-(V-50)}{RT/2F}}\right] \tag{39}$$

$$I_{Ca,\,L,\,K} = P_{CaK}P_{Ca}dff_2 \frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[\left[K^{+}\right]_i e^{\frac{50}{RT/F}} - \left[K^{+}\right]_o e^{\frac{-(V-50)}{RT/F}}\right] \tag{40}$$

$$I_{Ca,\,L,\,Na} = P_{CaNa}P_{Ca}dff_2 \frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[\left[Na^{+}\right]_i e^{\frac{50}{RT/F}} - \left[Na^{+}\right]_o e^{\frac{-(V-50)}{RT/F}}\right] \tag{41}$$

$$I_{Ca,\,L,\,Ca,\,ds} = 4P_{Ca}dff_{2ds}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/2F}}}\left[\left[Ca^{2+}\right]_i e^{\frac{50}{RT/2F}} - \left[Ca^{2+}\right]_o e^{\frac{-(V-50)}{RT/2F}}\right] \quad (42)$$

$$I_{Ca,\,L,\,K,\,ds} = P_{CaK}P_{Ca}dff_{2ds}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[\left[K^{+}\right]_i e^{\frac{50}{RT/F}} - \left[K^{+}\right]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad (43)$$

$$I_{Ca,\,L,\,Na,\,ds} = P_{CaNa}P_{Ca}dff_{2ds}\frac{\frac{V-50}{RT/F}}{1-e^{\frac{-(V-50)}{RT/F}}}\left[\left[Na^{+}\right]_i e^{\frac{50}{RT/F}} - \left[Na^{+}\right]_o e^{\frac{-(V-50)}{RT/F}}\right] \quad (44)$$

Note: P_{Ca} is the constant field permeability multiplied by F.

The Transient Outward Current

The equations for this current are missing from the Appendix in the paper.

They should read as follows:-

$$I_{to} = G_{to}\left(G_{tos} + S(1-G_{tos}))R\right)\frac{\left[Ca^{+}\right]_i}{\left[Ca^{+}\right]_i + k_{cato}}(V - E_K)$$

$$R_{ss} = \frac{1}{1+e^{-0.2(V+4)}}$$

$$\dot{R} = 333(R_{ss} - R)$$

$$\dot{S} = 0.033e^{-V/17}(1-S) - \frac{33}{1+e^{-(V+10)/8}}S$$

Default values are:- $G_{to} = 0.005$ µS, $G_{tos} = 0$, $k_{cato} = 0.0005$

Drug-Receptor Interactions

Equations 3a, 65 and 66 should read:-

$$\dot{h} = 20e^{-0.125(V+75)}(1-b-h) - \frac{2000}{1+320e^{-0.1(V+75)}}h + (l_r h_D(1-m^3) + l_a m^3 h_D) - (k_r h(1-m^3) + k_a m^3 h)[D] \tag{3a}$$

$$\dot{h}_D = 20e^{-0.125(V+75+\Delta V)}(1-h_D) - \frac{2000}{1+320e^{-0.1(V+75+\Delta V)}}h_D + (k_r h(1-m^3) + k_a m^3 h)[D] - (l_r h_D(1-m^3) + l_a m^3 h_D) \tag{65}$$

$$\dot{b} = (k_r h(1-m^3) + k_a m^3 h + k_i(1-b-h))[D] - (l_r h_D(1-m^3) + l_a m^3 h_D + l_i(b-h_D)) \tag{66}$$